

EDUCATION

Northwestern University

Ph.D Candidate, Transportation Systems Analysis and Planning
Master of Science, Transportation Systems Analysis and Planning
NSF Graduate Research Fellow

Evanston, IL
Sep. 2021 – Jun. 2026

Sep. 2023 – Present

Goshen College

Bachelor of Arts in Physics, minor in Mathematics. *Summa cum laude*.

Goshen, IN
Aug. 2015 – May 2019

PUBLICATIONS AND PRESENTATIONS

Journal Articles

- **Aeschliman, S.**, Pougala, J., & Stathopoulos, A. (2026), “Fully-Bayesian imputation of incomplete transit panel data using temporal Gaussian Processes: simulations and real-world evidence.” Manuscript under review.
- **Aeschliman, S.** & Stathopoulos, A. (2025), “Trade-offs in transit public safety policy: balancing enforcement and quality improvements.” *Transportation Research Part A: Policy and Practice*.
- **Aeschliman, S.**, Huang, E., Hofstra, A., & Stathopoulos, A. (2025), “Examining Riders’ Subjective Equity Standards for Transit Service: How will they shape future transit planning and operations?” *Transport Policy*.
- Said, M., **Aeschliman, S.**, & Stathopoulos, A. (2023), “Robots at your doorstep: acceptance of near-future technologies for automated parcel delivery.” *Scientific Reports*.
- Alam, Md.R., Hou, C., **Aeschliman, S.**, Zhou, Y., & Guo, Z. (2022), “Optimization-based trip chain emulation for electrified ride-sourcing charging demand analyses.” *Transportation Letters*.
- Zhou, Y., **Aeschliman, S.**, & Gohlke, D. (2021), “Affordability and household transportation fuel costs by region and socioeconomic factors.” *Transportation Research Record*.

Conference Papers & Presentations

- **Aeschliman, S.**, Pougala, J., & Stathopoulos, A. “Choice-survival models of transit panel survey participation.” *TRB 2026*, Washington, DC.
- **Aeschliman, S.** & Stathopoulos, A. (2025), “Fully-Bayesian imputation for transit panel survey data.” *Modeling Mobility 2025*, Minneapolis, MN.
- **Aeschliman, S.** & Stathopoulos, A. (2025), “Improving transit panel survey inferences: imputation, selection models, and survival analysis.” *ITEA 2025*, Evanston, IL.
- **Aeschliman, S.** & Stathopoulos, A. (2025), “Tracking public transit satisfaction with multilevel longitudinal models.” *TRB 2025*, Washington, DC.
- **Aeschliman, S.** & Stathopoulos, A. (2024), “Trade-offs in transit public safety policy: balancing enforcement and quality improvements.” *IATBR 2024*, Vienna.
- **Aeschliman, S.** & Stathopoulos, A. “The Return to Transit in a Changing Landscape: Telework and Safety Concerns.” *TRB 2024*, Washington, DC.
- Said, M., **Aeschliman, S.**, & Stathopoulos, A. “Robots at Your Doorstep? Acceptance of Near-Future Modes for Parcel Delivery.” *TRB 2023*, Washington, DC.
- **Aeschliman, S.**, Zhou, Y., Macal, C., & Zhou, Z., “Agent-based modeling of electric vehicles with time-of-use electricity rates”, *EVS 33*, Portland, OR.
- Hou, C., Guo, Z., **Aeschliman, S.**, Zhou, Y., Afifah, F., & Huang, J., “Energy impact analyses of electrified ride-sourcing services without trip chain data”, *TRB 2021*, Washington, DC.

Technical Reports

- Zhou, Y., Siddique, N., Mintz, M., **Aeschliman, S.**, & Macal, C. (2022). “Electric Vehicle and Infrastructure Systems Modeling in Washington D.C. and Baltimore” (ANL-22/28). Argonne National Lab. (ANL), Argonne, IL (United States); Exelon, Chicago, IL (United States).
- Zhou, Y., Mintz, M., Stephens, T., **Aeschliman, S.**, & Macal, C., “Electric Vehicle Adoption in Illinois”, Energy Systems Division, Argonne National Laboratory, June 2020.

Other Presentations

- “Simulating Forest Fires” (with Luke Rush), Goshen College Academic Symposium, March 15, 2019.

TEACHING

- Guest Lecture: Trip Generation Modeling in R** Winter 2025, Fall 2025, Winter 2026
Northwestern University, CEE 480: Travel Demand Modeling
- Guest Lecture: Behavior Modeling for Smart Cities and Technologies** Winter 2025
Northwestern University, CEE 201: Decision Science in the Age of Smart Technologies
- TA: Engineering Analysis II** Winter 2023
Northwestern University, McCormick School of Engineering
- Weekly recitations, office hours, and grading

PROFESSIONAL EXPERIENCE

- Systems Modeling and Software Engineer** Sep. 2019 – May 2020 (intern), May 2020 – Sep. 2021 (staff)
Argonne National Laboratory, Energy Systems Division Lemont, IL
- Updated and maintained the agent-based transportation energy analysis model (ATEAM), providing concrete EV charging station placement recommendations to ComEd and Exelon in Chicago and DC/Baltimore
 - Performed data analysis and visualization with Python and R on various transportation energy-related projects for government and corporate sponsors
 - Used gradient boosting and deep neural networks for predictions on vehicle miles traveled and freight mobility
 - Contributed to technical reports prepared by the group for public access

MAJOR AWARDS AND ACHIEVEMENTS

- National Science Foundation Graduate Research Fellow (Sep. 2023 - Present)
- NAIA Men's Soccer Academic All-American (2017, 2018)
- College College Presidential Leadership Award recipient (2015-2019)

INTERNSHIPS

- Science Undergraduate Laboratory Internship** Sep. 2019 – Apr. 2020
Argonne National Laboratory Lemont, IL
- Learned Java and its implementation in ATEAM
 - Devised and performed a case study on time-of-use electricity rates in ATEAM, which culminated in a conference paper submission and acceptance
 - Maintained Argonne's alternative fuel vehicle sales database on a monthly basis
- Maple Scholars Research Program** May 2017 – July 2017
Goshen College Goshen, IN
- Designed and built small, car battery-powered air quality sensors (for Ozone, CO, VOCs, and particulates) with a Raspberry Pi controller
 - Programmed the controller's data collection (including communication with sensors), power management, and GPS tracking using Python and C
 - Visualized Goshen-area spatial air quality data on a website using the Google Maps API